

Database Normalization Solutions

Step-by-Step Practical Implementations (1NF, 2NF, 3NF)

1. Normalization into 1st Normal Form (1NF)

Rule for 1NF: A relation is in 1NF if and only if the domain of each attribute contains only atomic (indivisible) values, and the value of each attribute contains only a single value from that domain.

Analysis: In the original table, the Hobbies column contains multi-valued comma-separated lists (e.g., "Cricket, Swimming, Football"). To achieve 1NF, these values must be broken down into atomic records, resulting in multiple rows per member id.

Normalized 1NF Table:

Member_Id	First_Name	Last_Name	Hobbies
101	Jayson	Mark	Cricket
101	Jayson	Mark	Swimming
101	Jayson	Mark	Football
102	Ram	Ganesh	Swimming
102	Ram	Ganesh	Running
102	Ram	Ganesh	Music
103	Raj	Kishore	Dancing
103	Raj	Kishore	Singing
103	Raj	Kishore	Running

2. Normalization into 2nd Normal Form (2NF)

Rule for 2NF: The relation must be in 1NF and there must be **no partial dependencies**. Every non-prime attribute must be fully functionally dependent on the primary key.

Analysis: The composite primary key for the table is (Empno, Training).

- Training_Date depends fully on both Empno and Training.
- Dept depends solely on Empno, which is only a proper subset of the composite key. This constitutes a **Partial Dependency**.

Solution: Decompose the table into two distinct relations to remove the partial dependency.

Table 1: Employee Department Details (Primary Key: Empno)

Empno	Dept
101	TT
101	BU
102	TT

Table 2: Training Records (Composite Primary Key: Empno, Training)

Empno	Training	Training_Date
101	Oracle SQL	12-Aug-2015
101	Java	21-Aug-2015
102	Oracle SQL	18-Sep-2014

3. Normalization into 3rd Normal Form (3NF)

Rule for 3NF: The relation must be in 2NF and there must be **no transitive dependencies**. A non-prime attribute must not depend on another non-prime attribute.

Analysis: Member_Id is the primary key. The dependencies present are: *Member_Id* → *Sports* and *Sports* → *Fees*. Consequently, *Member_Id* → *Fees* is a **Transitive Dependency** because Fees is determined by the sport, not directly by the unique member.

Solution: Separate the relation into a member mapping table and a standalone sports fee catalog.

Table 1: Member Profiles (Primary Key: Member_Id)

Member_Id	First_Name	Last_Name	Sports
101	Rajesh	Chand	Cricket
102	Jayesh	Raj	Hockey
103	Mark	Dorson	Football

Table 2: Sports Fees Structure (Primary Key: Sports)

Sports	Fees
Cricket	100
Hockey	80
Football	90